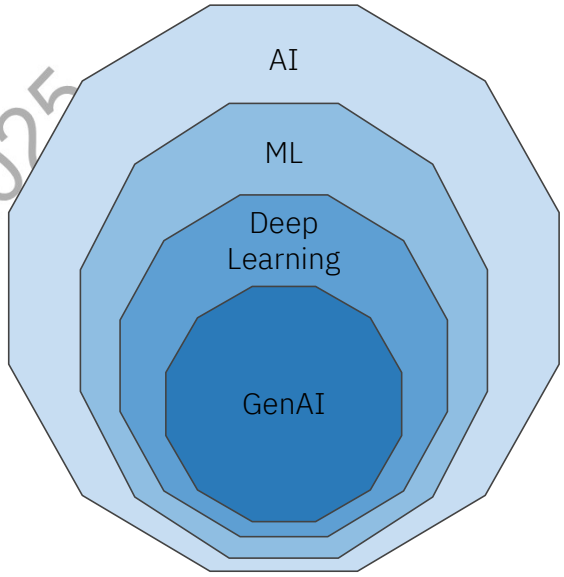
Introduction to AI

Week 8: Deep learning

Nguyen Tuan Khang *PhD.* | khangnt@vn.ibm.com |
090 830 6668
Data & AI Leader, IBM ASEAN

Tran Dang An
Assistant

USTH 2025



1

Objectives

- Limitation of DNN and how Deep Learning evolved
- Use of embeddings to transform raw data into meaningful vector space, support DNN to learn diverse inputs: text, images, audio, video
- CNNs for images
- Approach modern AI models to handle large input contents as the foundation to GenAI

USTH 2025

2

2

AI Hierarchy

Artificial Intelligence

W1: Basic concepts, history of AI
W2: Knowledge representation and logic
W3: Searching algorithms (pathfinding and text)

Machine Learning

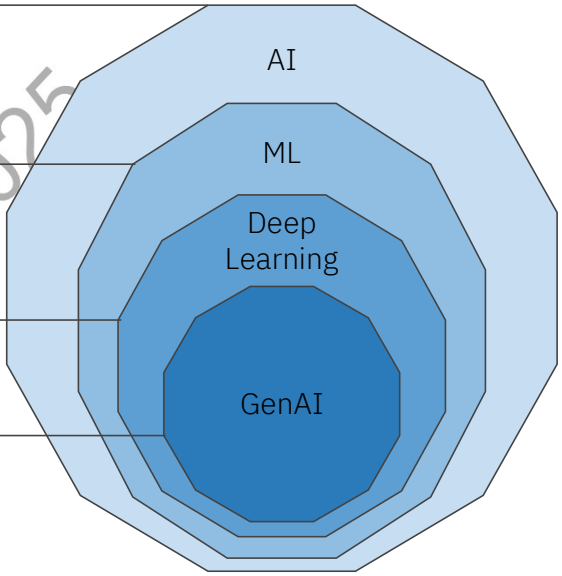
W4: Statistical learning
W5: Supervised machine learning
W6: Unsupervised machine learning

Deep Learning

W7: Perceptron learning and neural network learning
W8: Deep learning

Generative AI

W9 + W10: Modern AI and development in real world applications (LLM & GenAI)



USTH 2025

3

3

Contents

- From MLP to Deep Learning Families
- Embeddings for Deep Neural Networks
- Convolutional Neural Networks (CNN)

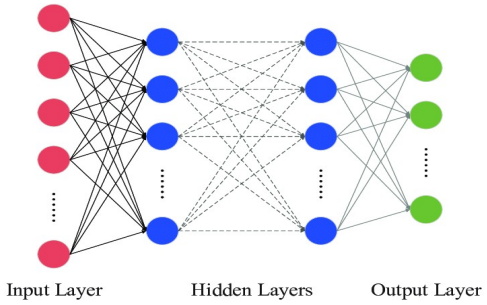
USTH 2025

4

4

From Deep MLP to Deep Learning Families

- Deep MLP with many hidden layers (DNN). The deeper the network, the more complex things it can understand.
- The DNN is good at tabular data
- What is about non-tabular data, e.g. text, image, video...?



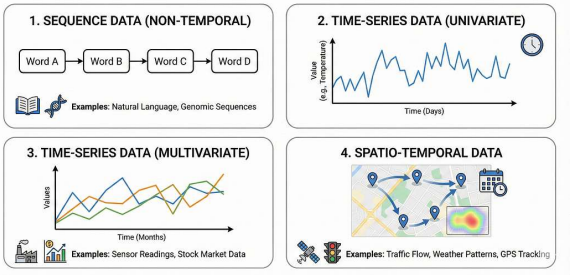
USTH 2025

5

5

Limitations of MLP (DNN)

- Learn naturally sequential or temporal data (language, time series)
- Handle spatial structure (e.g. images)
  - Images must be flattened (Matrix -> Vector).
  - Impact: neighboring pixels lose their context (the model forgets that "pixel A" is next to "pixel B").
- Scale to support parameter explosion
  - DNN require massive amounts of weights for large inputs.
  - Ex: A 200x200 image into 1,000 neurons = 40 Million parameters, too slow



USTH 2025

6

6

Models that can learn images and texts?

- Example that we need recognize a real image (like cat/dog or face) with size 200x200 even bigger like 4k images?
  - Flatten it to vector 40k data inputs results a complicated DNN
- How about models that can support natural language (e.g., using words or long sentences)?



Is any model that can learn complex data inputs, like the matrix (grid data), natural languages, time series or graph?

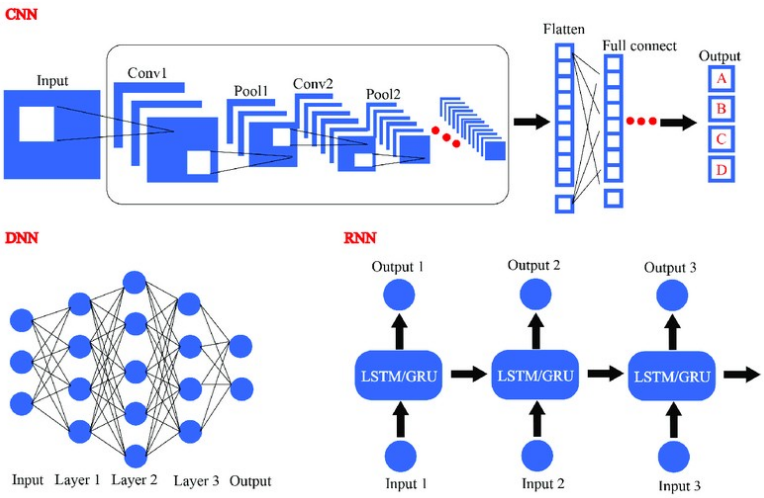
USTH 2025

7

7

DL Families for different purposes

- **DNN (Deep Neural Network):** Deep MLP, good for tabular data
- **CNN (Convolutional Neural Network):** Specialized for images
- **RNN:** Specialized for sequence data (Text, Audio, Time-Series). It remembers the past to predict the future.
- **Transformers:** The foundation for all LLM (GPT, Llama, Claude, ...)



USTH

8

8

Contents

- From MLP to Deep Learning Families
- Embeddings for Deep Neural Networks
- Convolutional Neural Networks (CNN)

USTH 2025

9

9

Quiz 1

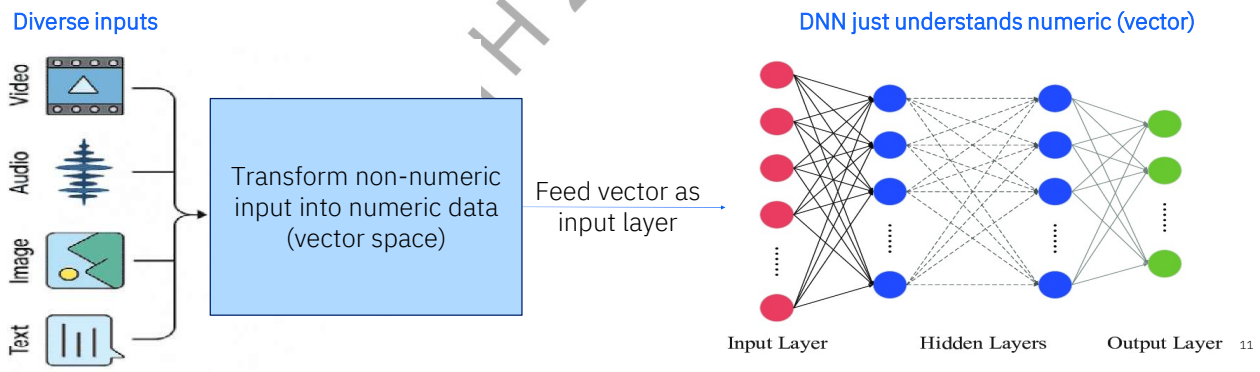
- Enhance DNN to learn diverse inputs (language, image, video, graph...)?



10

Text Meaning to Machine

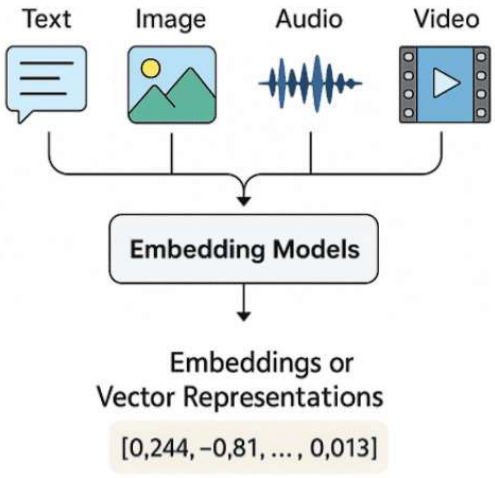
- Problem
  - DNN are mathematical engines, only understand numbers, vectors. DNN can not “read” text
- Solution
  - We must convert raw inputs (strings, images) into a numerical format that preserves meaning.



11

Embeddings Definition

Embedding refers to a representation learning technique that maps complex, high-dimensional data into a lower-dimensional vector space of numerical vectors.



12

Key Embedding Techniques

Techniques	Sample models	Type	Notes
Classical word embeddings	One-hot encoding Bag of Words (BoW)	Labeling Frequency-based	Sparse, no semantic meaning
Word embeddings	Word2Vec	Prediction-based (a pre-trained model)	Treat words in sentence separately (static)
Contextual embeddings	BERT GPT embeddings	Transformer-based (a pre-trained model)	Treat words within the context of the whole sentence (dynamic context) This is the modern standard.
Document embeddings	Doc2Vec S-BERT		Optimized for similarity, clustering, and retrieval.
Multimodal embeddings	Media embeddings (image, audio, video)		
Others	GNN embeddings Node2Vec		

13

13

Bag of Words (BoW, Frequency Based)

1. Build a vocabulary to contains all unique words in your dataset (consider as a dictionary)

2. Counting
   - For every sentence, create a vector where each slot corresponds to a word in the vocabulary.
   - The value is the count of how often that word appears.

3. Result
   - A sentence becomes a list of numbers representing word frequencies, we use this list as a vector

USTH 2025

	the	red	dog	cat	eats	food
1. the red dog →	1	1	1	0	0	0
2. cat eats dog →	0	0	1	1	1	0
3. dog eats food →	0	0	1	0	1	1
4. red cat eats →	0	1	0	1	1	0

14

14

BoW Examples

- Case study
  - Input
    - Doc 1: "This is the first document.“
    - Doc 2: "This document is the second document.“
    - Doc 3: "And this is the third one.“
    - Doc 4: "Is this the first document?"
  - Vocabulary: ['and' 'document' 'first' 'is' 'one' 'second' 'the' 'third' 'this’]

- Limitations
  - Sparsity: If you have 10,000 words, you may get a vector with 9,995 zeros.
  - No order: "Man bites dog" and "Dog bites man" produce the exact same vector.

BoW Matrix:
[0 1 1 1 0 0 1 0 1]
[0 2 0 1 0 1 1 0 1]
[1 0 0 1 1 0 1 1 1]
[0 1 1 1 0 0 1 0 1]

Document D1	*The child makes the dog happy* the: 2, dog: 1, makes: 1, child: 1, happy: 1
Document D2	*The dog makes the child happy* the: 2, child: 1, makes: 1, dog: 1, happy: 1

	child	dog	happy	makes	the	BoW Vector representations
D1	1	1	1	1	2	[1,1,1,1,2]
D2	1	1	1	1	2	[1,1,1,1,2]

15

15

Lab 8.1 BoW for Emojis

- Assume we have emojis as this table

Happy	😄	😘	🥰
Sad	😫	😒	😛
Excited	😄	🥳	😘
Sick	😫	🤮	🤒

- Embed the emoji vectors for these emojis?
  - Guidelines: Vocabulary (BoW): 4 words [Happy, Sad, Excited, Sick]

😘 = [...]
😫 = [...]
🤒 = [...]

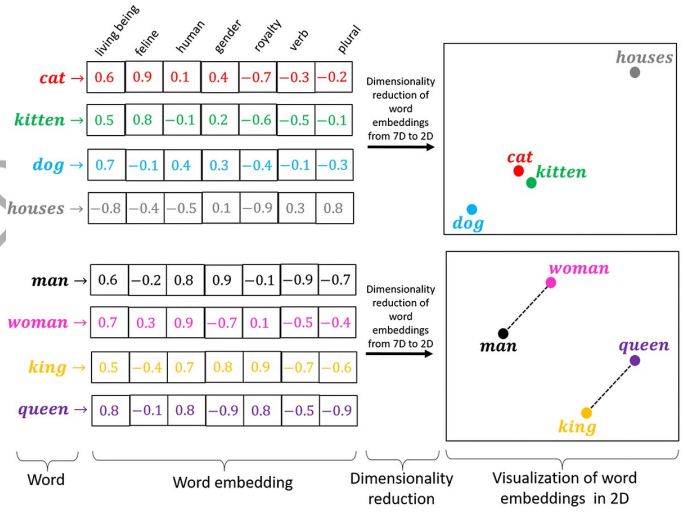
😘 = [1,0,1,0]
😫 = [0,1,0,1]
🤒 = [0,1,0,1]

16

16

Word2Vec (Prediction Based)

- Instead of counting words, we train a neural network to predict them.
- The network learns to group similar words together in vector space.
  - E.g "Car" and "Truck" end up with very similar coordinates because they appear in similar contexts

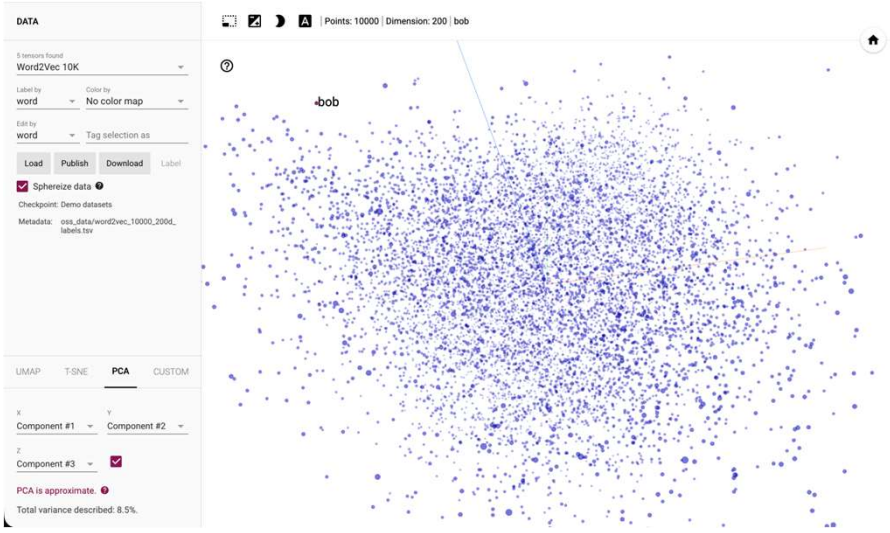


USTH 20

17

17

Word2Vec Embeddings Visualized



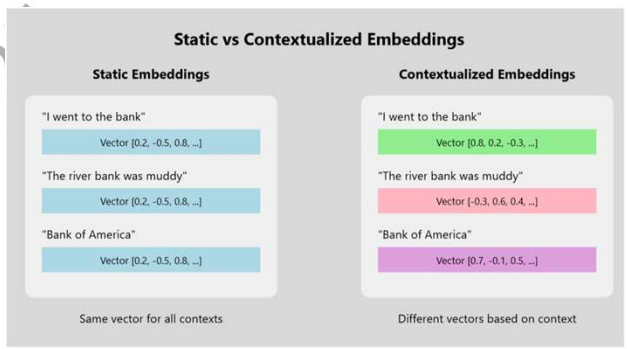
https://projector.tensorflow.org/

18

18

GPT (Contextualized)

- Word2Vec has a limit as it is static
  - The word **"Bank"** has the **exact same vector** in **"I went to the bank"** and **"Bank of America"**. The model is confused by the double meaning.
- Contextualized Embeddings
  - Models like GPT (Generative Pre-trained Transformer) generate embeddings dynamically.
  - *The vector for "Bank" changes based on the other words in the sentence.*
- We will research more about GPT in next lesson
  - Understand how GPT does this dynamic embeddings.
  - Understand the **Transformer architecture**.



USTH 202

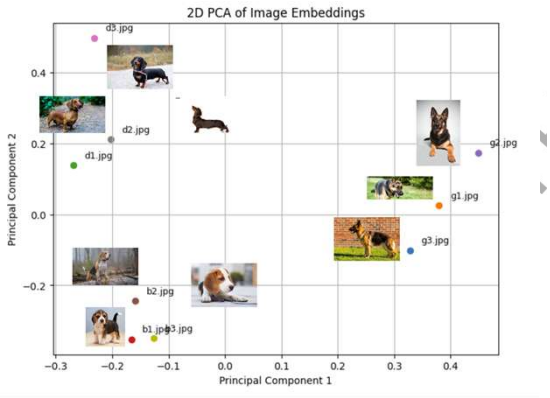
19

19

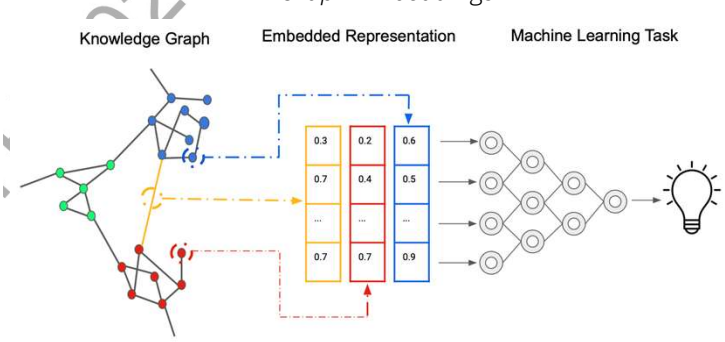
Other Embeddings

- It's not just for text, we can embed *anything* into vectors

Image Embeddings



Graph Embeddings



025

20

20

Key Takeaways

- DNN works with tabular data (numeric)
- If data input is non-tabular like words, sentences, images… we need transform them into vector (called embeddings)
  - We can embed words, sentences
  - We can embed images, different from flatten, even both produce vector inputs

USTH 2025

21

21

Contents

- From MLP to Deep Learning Families
- Embeddings for Deep Neural Networks
- Convolutional Neural Networks (CNN)

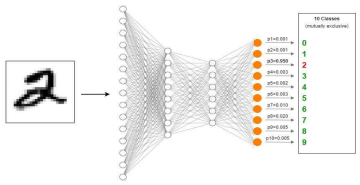
USTH 2025

22

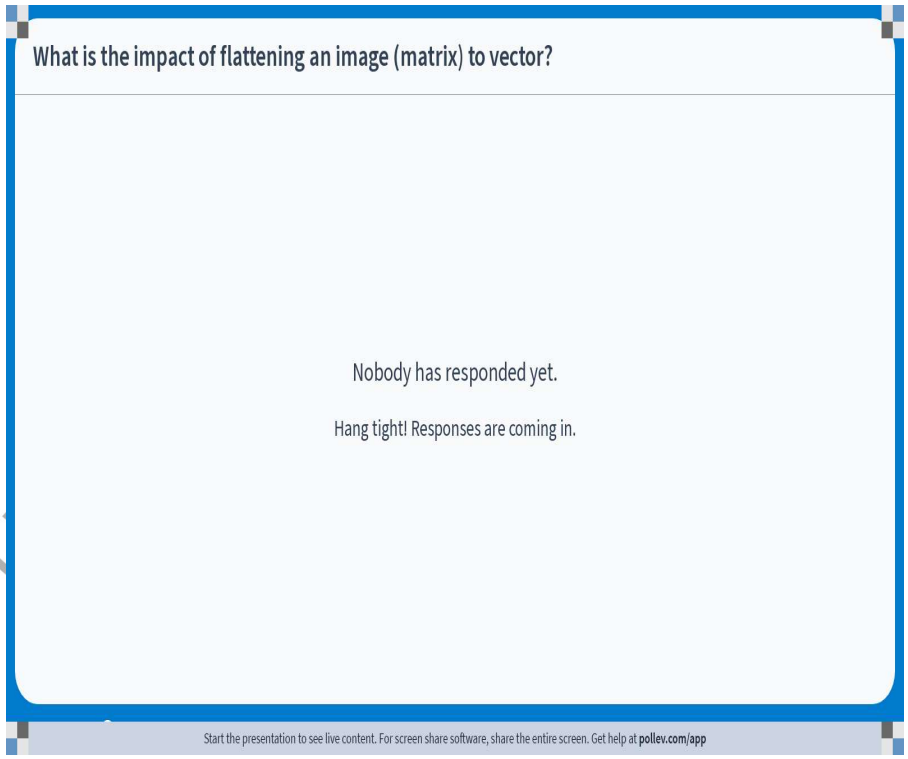
22

Quiz 2

- MLP to recognize hand-writing using size 28x28
  - Flatten the image to vector 784 data inputs



- What happens if the image is 1024x1024 or 4k image?

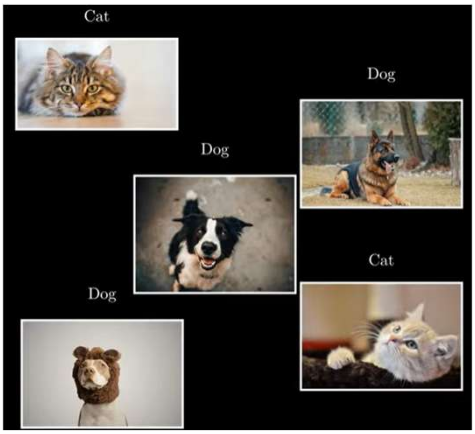


23

DL Model for Images

- By transforming image into vector (either flatten or embedding), we might destroy the shapes (circle, lines)

We need an AI model that can "see" grids



USTH 2025

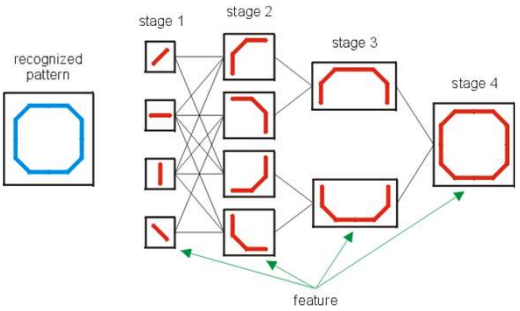
24

24

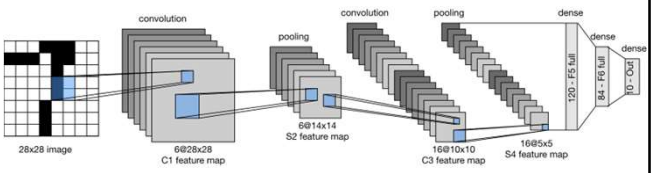
History of DL for Images

- 1980: The Spark (Fukushima)
  - Inspired by the biological cat's eye, the Neocognitron introduced the idea of layers (**Convolution** & **Pooling**)
  - But no Backpropagation to learn.
- 1989-1998: The Breakthrough (Yann LeCun)
  - LeCun applied **backpropagation** to this architecture at Bell Labs.
  - LeNet-5: The first successful CNN, designed to read handwritten digits (MNIST).
- Impact
  - In the late 90s, LeNet-5 automated ATM check reading for 10-20% of all US banking checks.
  - Note: Despite this success, CNNs hibernated until 2012 (AlexNet) due to hardware limitations.

Neocognitron



LeNet-5



USTH 2025

25

25

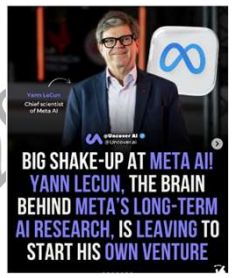
Yann LeCun



2013
(joined as Chief AI Scientist)



June 2025



Nov 2025



Jan 2026

- 1987: Refined the backpropagation
- 1998: Invented CNN
- Joined Meta in 2010 and leaved in 2025
- 2024: VinFuture Prize and TIME100 Impact

26

26

Quiz and Approach

- Flatten is not suitable for large images
  - Find a technique to resize the image to smaller size before flattening
- How to normalize an image to sharpen or blur it?
  - Normalize pixels using min/max/mean functions from its neighbors
- How to resize an image that minimize the loss of information?
  - Convert multiple pixels into 1 pixel

USTH 2025

27

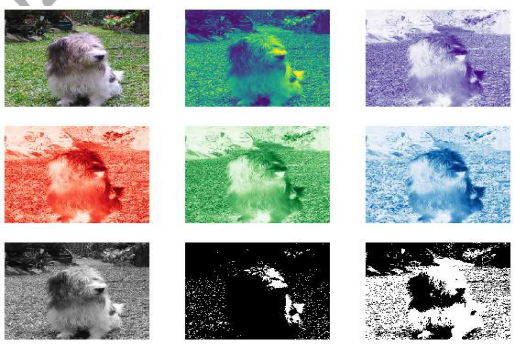
27

Image Filter

Filters for Camara



What machine sees



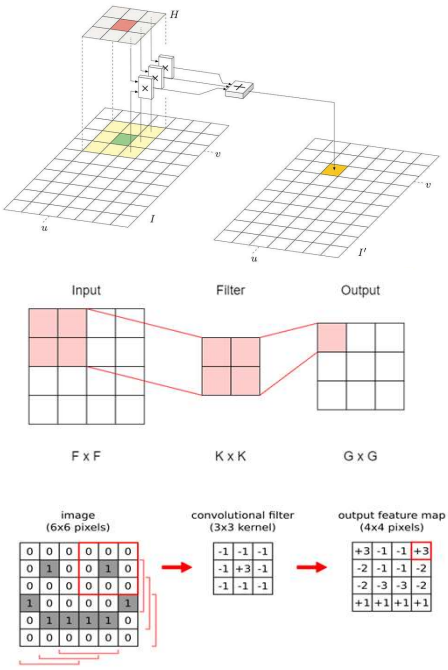
USTH 2025

28

28

Image Filter in explained

- Replace every pixel by the average of its neighbor pixels, can keep the same size or resize it smaller
- Filter is also called as Kernel
- Filter matrix
  - Odd matrix: 3x3, 5x5
  - Even matrix: 2x2, 4x4

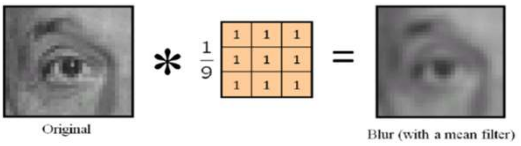


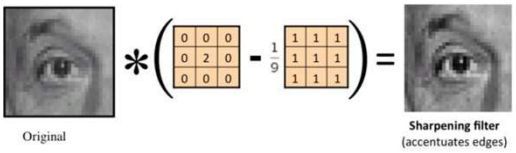
USTH 2025

29

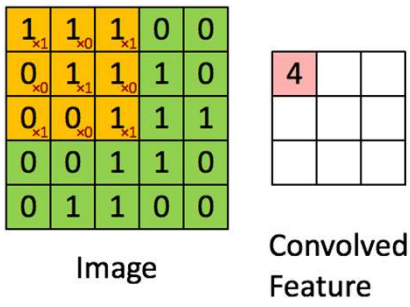
29

Examples





Blur or Sharpen an image
Keep the size unchanged



Other technique: averaging
Resize to smaller, also called as Convolution

Replace each pixel by the average of its neighbors. This assumes that neighboring pixels are similar and the noise is independent across pixels.

30

30

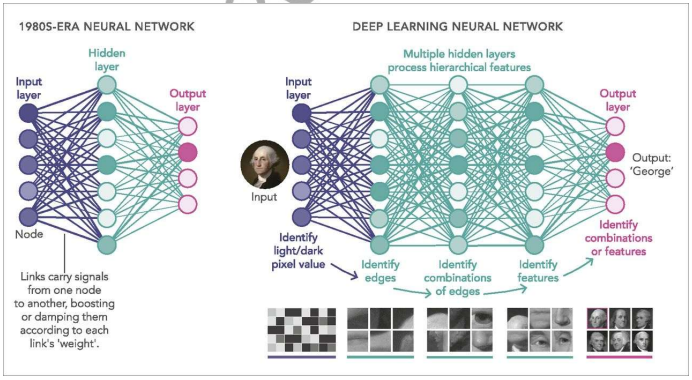
CNN for Images

DNN and its problem

- Parameter explosion: A standard DNN on a 1000x1000 image requires 1 billion weights for just the first layer
- Lost structure: Flattening an image into a list destroys spatial context (e.g., nose is above mouth).

CNN and its strength

- Parameter sharing: Uses the same small "filter" across the whole image.
- Spatial invariance: Recognizes objects (e.g., a cat) regardless of where they are in the frame.

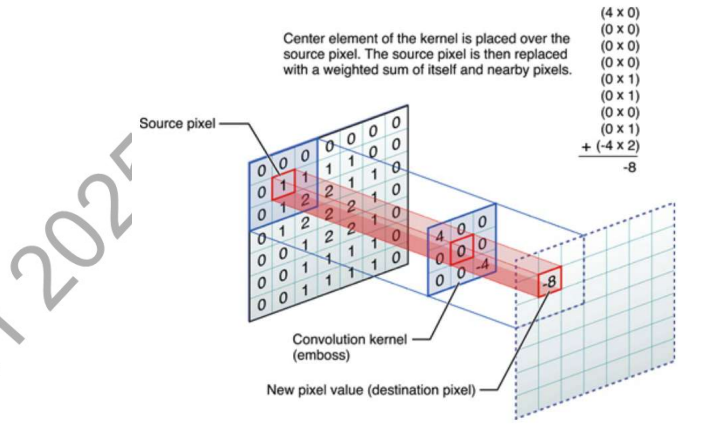


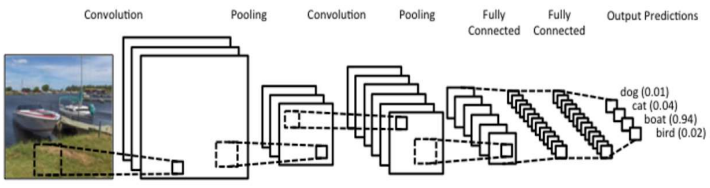
31

31

CNN Highlights

- Designed specifically to process data that has a grid-like topology (e.g images).
- Instead of looking at every pixel at once, CNN uses a "Filter" to scan across the image to detect patterns. It starts by finding edges, then shapes, then textures.

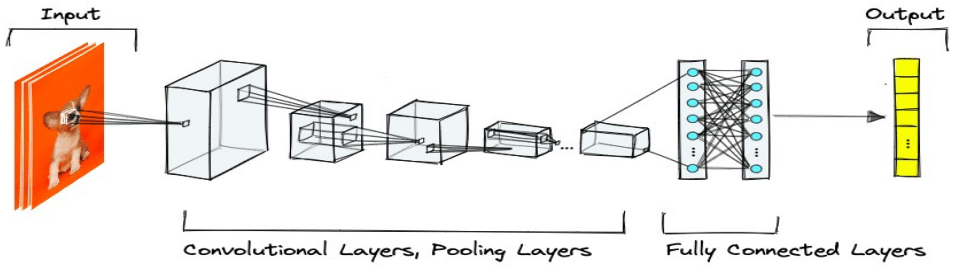




32

32

CNN Layers



Convolutional Layer

- Use filer over the image to detect features.
- Deeper layers recognize complex objects, features.

Pooling Layer

- Use filter to resize image (e.g. summarizing small regions into 1 pixel)
- Reduces computation costs.

Fully connected (DNN)

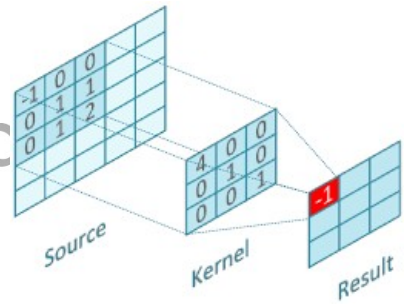
- Flattens 2D feature maps into a 1D vector.
- Performs the final decision. Combines features to output a class label (e.g., "Dog").

33

33

Convolution Layer

- The Filter (Kernel)
  - A small matrix (e.g., 3x3 pixels) that acts as a feature detector (e.g., finds vertical lines).
- The Operation
  1. Slide: The filter moves across the image.
  2. Multiply & Sum: Calculates the dot product at each step.
  3. Output: Creates a feature map highlighting where the feature exists.

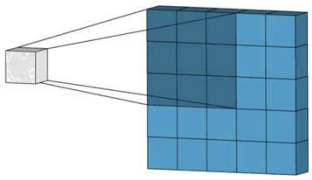


Original	Gaussian Blur	Sharpen	Edge Detection
$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$
			

34

34

Lab 8.2: Apply the Filter “Mean”



Original image pixel values

[0. 0. 0. 0. 0. 0. 0.]
[0. 0. 0. 0. 0. 0. 0.]
[0. 0. 1. 1. 1. 0. 0.]
[0. 0. 1. 1. 1. 0. 0.]
[0. 0. 1. 1. 1. 0. 0.]
[0. 0. 0. 0. 0. 0. 0.]
[0. 0. 0. 0. 0. 0. 0.]

Filter (3x3)

[0.11 0.11 0.11]
[0.11 0.11 0.11]
[0.11 0.11 0.11]

Filtered image pixel values

[0.11 0.22 0.33 0.22 0.11]
[0.22 0.44 0.67 0.44 0.22]
[0.33 0.67 1. 0.67 0.33]
[0.22 0.44 0.67 0.44 0.22]
[0.11 0.22 0.33 0.22 0.11]

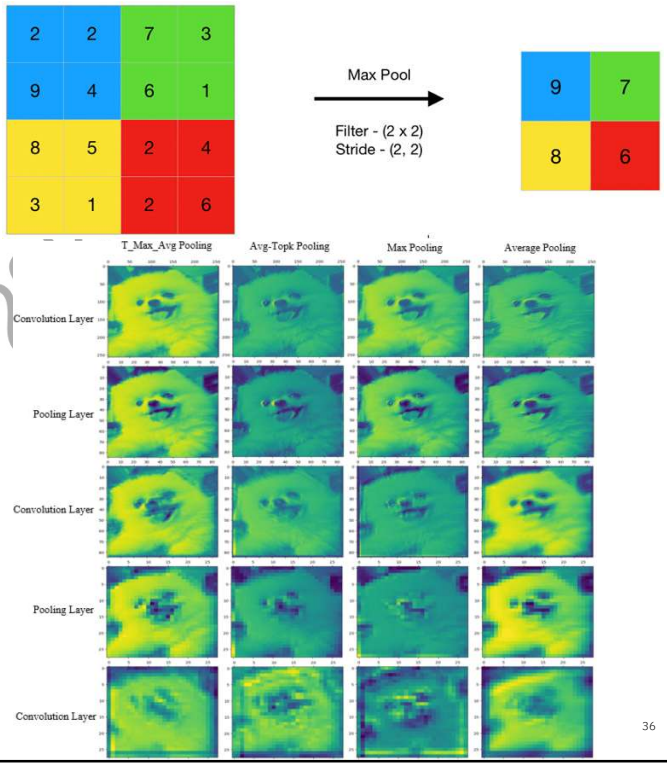
USTH

35

35

Pooling Layer

- Drastically reduce image dimensions and computation. (Downsampling)
- The Filter
  - Slide a filter (e.g., 2x2) over the map and summarizes the data.
  - Use Max pooling filter to select only the highest value in the window. It preserves the strongest feature (e.g., the sharpest edge) while discarding noise.
- Benefit
  - Makes the network invariant to small shifts or distortions (e.g., the object moves slightly, but the max value remains).



USTH 2

36

36

Filter: Conv layer vs Pooling layer

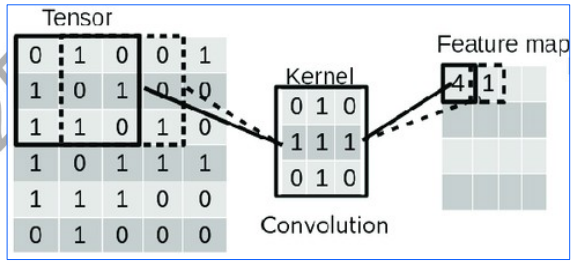
Factor	Convolution Filer	Pooling Filter
Purpose	Feature extraction	Dimensionality reduction
Params	Learnable weights	Fixed weights
Size	Odd, normally n=3 (3x3) N can be 5x5	Even, normally n=2 (2x2) N can be 4x4 for 4k images
Stride	Normally 1	Normally 2
Function	Element-wise multiple and sum (dot product)	Max, avg
Outcomes	Create detailed features	Reduce spatial size of features
Size of outcomes	N-2	N/2

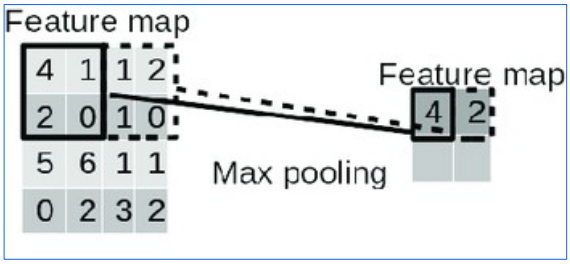
37

37

Choosing the Filter sizes (Kernel)

- Standard: 3 x 3 (odd) – conv layer
  - Most popular: It is the smallest size that captures a "center" pixel and all its immediate neighbors.
  - Efficiency: Stacking two 3 x 3 layers is better than one 5 x 5 layer.
  - Default choice for almost all modern architectures (VGG, ResNet, ...).
- Rare: 2 x 2, 4 x 4 (even) – pooling layer
  - Use for Pooling: 2 x 2 is the standard size for Pooling layers
  - Avoid for Conv : Even numbers don't have a single "center" pixel, which causes object shifting issue.
- Large: 5 x 5 or 7 x 7
  - Sometimes used only in the very first layer if the input image is massive (e.g., 4K resolution) *to reduce image's dimensions aggressively* at the start.





USTH 2025

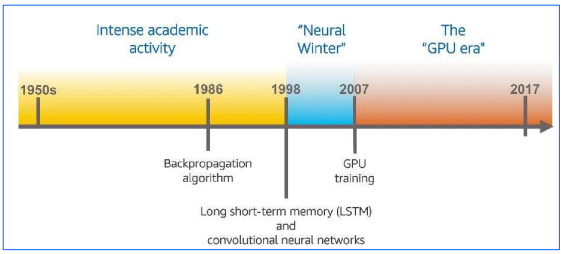
38

38

GPU for Deep Learning

- 1999: Nvidia introduced the first GPU GeForce 256 for graphics rendering.
  - Graphics problems are vector and matrix operations, such filtering images or transforming 3D using matrix multiplications (mil of vectors in parallel)
  - Remind that CNN was introduced in 1998
- 2000s: GPU became programable
  - Matrix - matrix and matrix - vector multiplications are what GPUs do best
  - These operations are also the core computation in neural networks
- 2010s onward: GPUs became central to deep learning

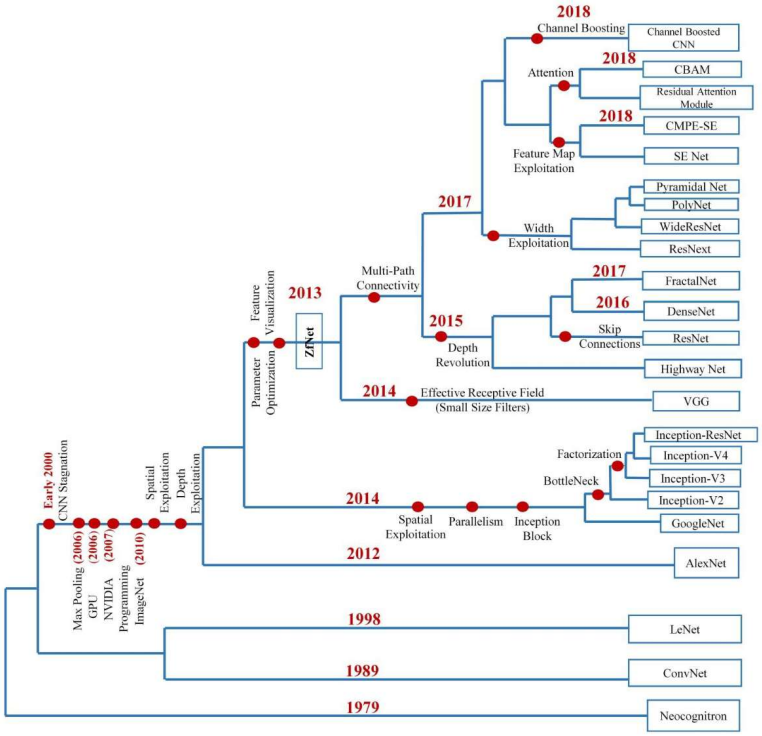
USTH 2025



39

39

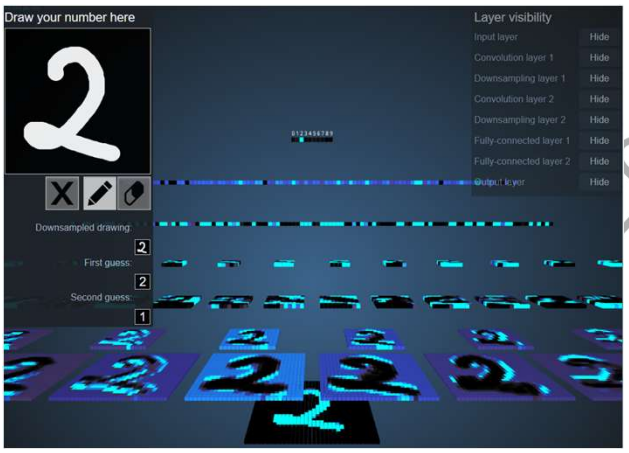
CNN Families

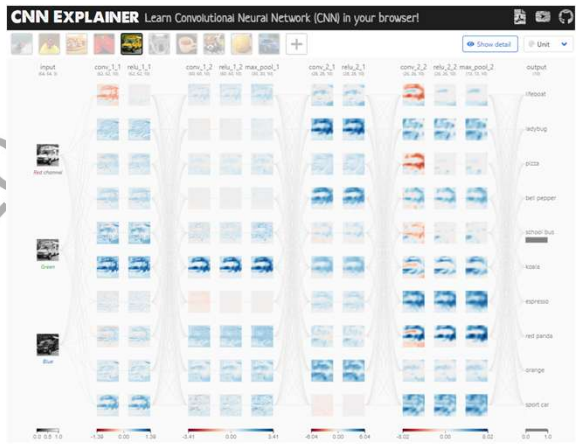


40

40

CNN Simulation





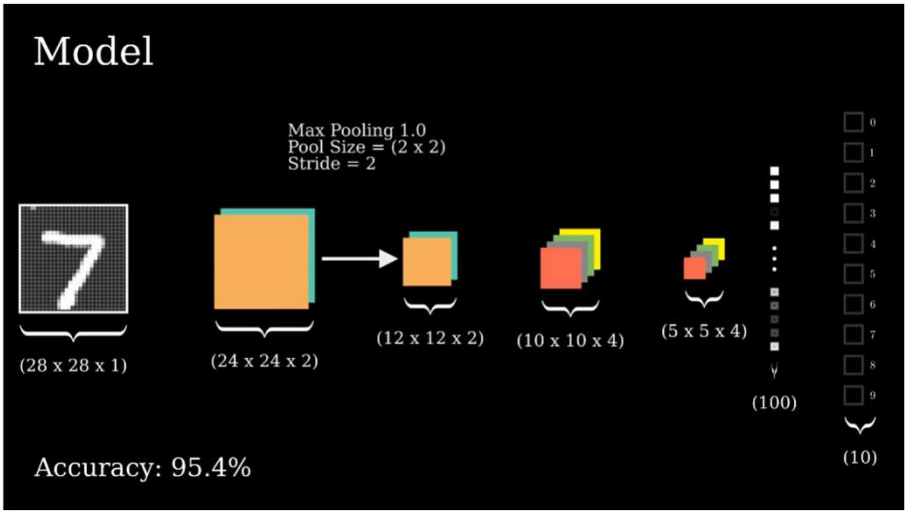
https://adamharley.com/nn_vis/cnn/3d.html

https://poloclub.github.io/cnn-explainer/

41

41

CNN Video



https://www.youtube.com/watch?v=JboZfxUjLSk&t=218s (6m)

42

42

Key Takeaways about CNN

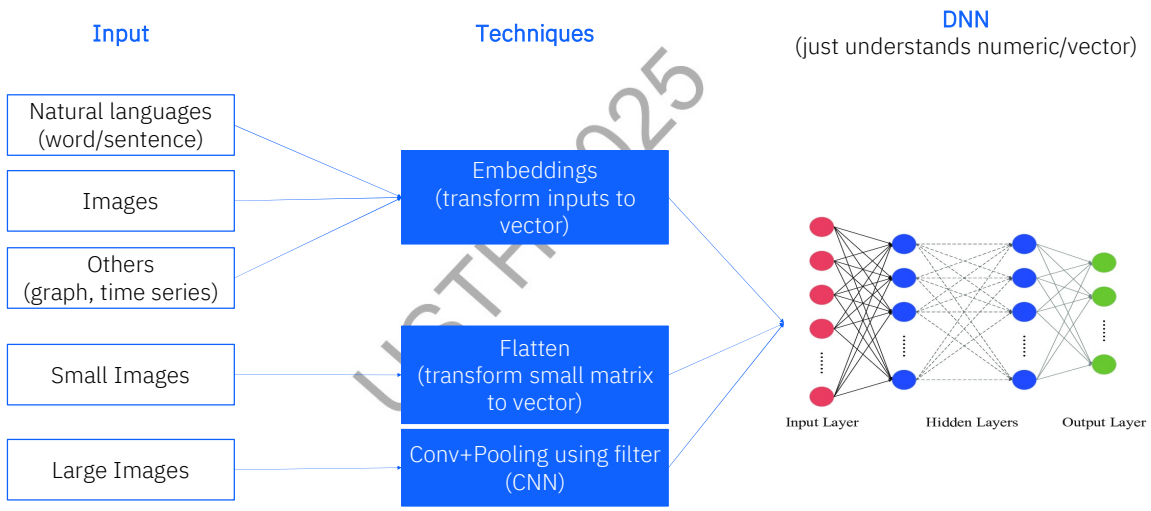
- Best for: Images & video like FaceID, medical scans.
- Pros: Scans like a flashlight, understands shapes, ignores object position.
- Cons: Heavy compute as it requires powerful GPUs

USTH 2025

43

43

Summary for DL



USTH 2025

44

44

Thank you!

USTH 2025

45

45

USTH 2025